॥ श्रीगणेशाय नमः ॥

Ages

Q. Arun's age after 15 years will be 5 times his age 5 yrs back. What's his current age?

→ $x+15 = 5(x-5)$

$40 = 4x$

$x = 10$

Q. The ages of 2 persons differ by 16 years. If 6 years ago, the elder was 3 times as old as the younger, find their current ages?

→ $x, x+16$

$(x+16-6) = 3(x-6)$

$10 = 2x-18$

$2x = 28$

$x = 14 \quad x+16 = 30$

Q. The product of the ages of A and B is 240. If twice the age of B is more than A's age by 4 years, what is B's age?

→ $x \quad \frac{240}{x}$

A B

$\frac{480}{x} = x + 4$

$x - \frac{480}{x} + 4 = 0$

$x^2 - 480 + 4x = 0$

$x^2 + 4x - 480 = 0$

$x^2 + 24x - 20x - 480 = 0$

$x(x+24) - 20(x+24) = 0$

$x = 20 \quad \frac{240}{x} = \frac{240}{20} = 12$

12

Q. A person was asked to state his age in years. His reply was "Take my age three years hence, multiply it by 3, subtract 3 times my age three years ago, and you'll know how old I am".

→ $3(x+3) - 3(x-3) = x$

$3x + 9 - 3x + 9 = x$

$x = 18$

Q. A father said to his son, "I was as old as you're rn at the time of your birth". If the father's age is 38 years now, the son's age 5 years back is ___.

→ $38 - x = x$

$x = 19$

$x - 5 = 14$

Q. My brother is 3 years elder to me. My father was 28 when my sister was born. My mother was 26 when I was born. If my sister was 4 years of age when my brother was born, then what was the age of my father and mother when my brother was born?

→ my age = x.

brother = $x+3$

sister = $x+7$

father = $x+7+28 = x+3+32$ = brother + 32

mother = $x+26 = x+3+23$ = brother + 23

Q. If 6 years are subtracted from the present age of G and the remainder is divided by 18, then the present age of his grandson A is obtained. If A is 2 years younger to M whose age is 5, then what's the age of G?

→ $\frac{(n-6)}{18} = 3$

$n = 54+6 = 60$

Q. Q is as much younger than R as he is older than T. If the sum of the ages of R and T is 50, what is definitely the difference between R and Q's age?

→ $(R-Q) = (Q-T)$

$R+T = 2Q = 50$

$Q = 25$

CANNOT BE DETERMINED

Q. The total age of A and B is 12 years more than the total of B and C. C is ___ years younger than A.

→ $A+B = 12+B+C$

$C = A-12$

12

=

Q. Ten years ago, the ages of the members of a joint family of 8 people added up to 231 years. 3 years later, 1 of them died at the age of 60. A child was born during the same year. After 3 more years, one more member died at 60 and another child was born. The current average age of the 8-member family is nearest to ____.

→

10 yrs ago, total = 231

After 3 yrs → total = $231 + 8(3) - 60 = 195$

After 3 more years → total = $195 + 8(3) - 60 = 159$

Now 4 more years to present. → total = $159 + 4(8) = 191$

avg = $\frac{191}{8} = 23.875 = 24$

Q. The avg age of a couple is 24 years. After their twins were born, the average age of the family became 13.5. The average age of the family just after the 3rd child was born was 13.2. The average age of the family after their 4th child was born was 16 years. The current avg age of the family is 19 years. What is the current age of the twins?

→ Say father = mother = 24

After twins, avg age = $\frac{(24+x)+(24+x)+0+0}{4} = 13.5$

$48+2x = 54$

$x = 3$

so twins were born when parents are 27, 27.

Say 3rd child is born after K years.

$\frac{27+K+27+K+0+K+0+K+0}{5} = 13.2$

$54+4K = 66$

$K = 3$

So now ages are 30, 30, 3, 3, 0

After y more years, another is born.

$\frac{66+5y}{6} = 16$

$5y = 96 - 30$

$y = 6$

Now ages are 36, 36, 9, 9, 6, 0

curr avg = 19.

$\frac{36+36+9+9+6+0+6(p)}{6} = 19$

$p = 3$

So now ages are 39, 39, 12, 12, 9, 3.

so ages of twins: 12 yrs each.

Q. The ratio of the present ages of P and Q is 5:7. If the difference between Q's present age and P's age after 6 years is 2, what are their current ages?

→ $5x, 7x$
P Q

$5x + 6 - 7x = 2$
$x = 2$
10, 14

$7x - (5x + 6) = 2$
$2x = 8$
$x = 4$
20, 28

Q. Sneha's age is 1/6th her father's age. Her father's age will be twice that of Vimal's age after 10 years. If Vimal's 8th b'day was celebrated 2 years ago, what is Sneha's current age?

→ $f = 6x$

$6x + 10 = 2(v + 10)$
$6x + 10 = 2v + 10 + 10$
$6x = 20 + 10 = 30$
$x = 5$

$v = 8 + 2 = 10$